1.

Answer: B Analysis: The rusting of an iron pot produces a new substance Fe2O3, which is a chemical change; the rest are physical changes.

2.

Answer: D Analysis: Chalcopyrite is CuSO4,5H2O; caustic soda NaOH, quicklime CaO, and soda ash Na2CO3.

3.

Answer: D Analysis: CH3COCH3 is acetone, and acetaldehyde is CH3CH2CHO.

4.

Answer: A Analysis: 2 ammonium ions are 2NH4+; ferrous chloride FeCl2, helium gas He, and 4 sulfuric acid molecules 4H2SO4.

5.

Answer: C Analysis: Dry ice (CO2) is a compound; liquid oxygen and mercury are elements, while hydrochloric acid is a mixture.

6.

Answer: C Analysis: O2 and O3 are different elements formed by the same element, and are allotropes of each other.

7.

Answer: D Analysis: In K2Cr2O7, K has a +1 oxidation state and O has a-2 oxidation state, so Cr has a +6 oxidation state.

8.

Answer: A Analysis: Milk is an emulsion; sugar water and iodine tincture are solutions, while mud water is a suspension.

9.

Answer: A Analysis: The yellow color of the combustion reaction corresponds to the element sodium.

10.

Answer: C Analysis: NaCl is a strong electrolyte; acetic acid and ammonia water are weak electrolytes, and sucrose is a non-electrolyte.

11.

Answer: B Analysis: CO2 dissolves in water to form H2CO3, which is conductive, but CO2 itself is a non-electrolyte.

12.

Answer: B Analysis: CO+CuO=Cu+CO2 does not belong to the four fundamental reactions, but it involves a change in oxidation state and is a redox reaction.

13.

Answer: C Analysis: Cl2 is easy to obtain electrons in the reaction and is commonly used as an oxidant.

14.

Answer: C Analysis: ^{3}sCl and ^{37}Cl have the same proton number but different neutron numbers, making them isotopes.

15.

Answer: B Analysis: The hydrolysis of Na2CO3 is alkaline with pH>7; KCl is neutral, while NH4Cl and Al2(SO4)3 are acidic.

16.

Answer: A Analysis: The purple litmus turns blue, indicating that the solution must be alkaline.

17.

Answer: A Analysis: H2 is insoluble in water and has a density lower than air, so it can be collected by drainage method + downward air displacement method.

18.

Answer: B Analysis: n (Na+) = 0.1 × 0.2 × 2 = 0.04 mol, the volume after mixing is about 0.5 L, c = 0.04/0.5 = 0.08 mol/L.

19.

Answer: C Analysis: Formaldehyde is toxic and cannot be used to soak seafood for preservation.

20.

Answer: B Analysis: Se is element 34, located in the fourth period of group VIA.

21.

Answer: B Analysis: Concentrated sulfuric acid absorbs water, increasing the solvent and decreasing the solute mass fraction.

22.

Answer: D Analysis: CH3OH and CH3CH2OH have similar structures with one CH2 difference, and are homologous.

23.

Answer: B Analysis: Benzene and liquid bromine undergo substitution reaction under Fe catalysis to form bromobenzene.

24.

Answer: B Analysis: Chemical equilibrium is a dynamic equilibrium in which the concentrations of each substance no longer change.

25.

Answer: C Analysis: SO3 reacts with alkali to form salt and water, which belongs to acidic oxides.

26.

Answer: B Analysis: Catalysts accelerate reaction rates by reducing reaction activation energy, and the mass and chemical properties remain unchanged before and after the reaction.

27.

Answer: C Analysis: Na2SO4 contains both ionic bonds and covalent bonds, while KCl contains only ionic bonds, so the types of chemical bonds are not entirely the same.

28.

Answer: B Analysis: CO_3^{2-} reacts with H+ to form CO2, making the clear lime water turbid.

29.

Answer: C Analysis: Zn+H2SO4=ZnSO4+H2 ↑, which is a displacement reaction.

30.

Answer: C Analysis: Na, Mg, and Al are all located in the third period.

31.

Answer: D Analysis: The atomic radius of elements in the same main group increases from top to bottom, so K>Na>Li.

32.

Answer: Electron configuration of chlorine atom: 2,8,7 Analysis: Cl is element 17, and its outer electron configuration is 2,8,7.

33.

Answer: B Analysis: 0.1 mol/L hydrochloric acid diluted 10-fold, concentration 0.01 mol/L, pH=2.

34.

Answer: C Analysis: v (NH3) = (0.8-0.4)/4 = 0.1 mol/(L·s), v (H2) = 3/2 v (NH3) = 0.15 mol/(L·s).

35.

Answer: A Analysis: To test SO_4^{2-}-first use dilute hydrochloric acid to eliminate interference, then add BaCl2 solution.

36.

Answer: No correct option Analysis: Temperature increase moves toward endothermic direction, while pressure increase moves toward direction of reduced gas molecule number. There is no option that satisfies both conditions simultaneously.

37.

Answer: C Analysis: Ethylene and HCl add to form chloroethane.

38.

Answer: B Analysis: Organic matter must contain carbon elements, and the rest of the statements are one-sided.

39.

Answer: B Analysis: The generation of 3 mol S (96 g) consumes 2 mol H2S.

40.

Answer: C Analysis: In AlCl3, $c(Cl^-) = 0.5 \times 3 = 1.5$ mol/L, which has the highest concentration.

41.

Answer: B Analysis: Endothermic reactions move forward as temperature decreases, while increasing gas molecular numbers move forward as reaction pressure increases.

42.

Answer: C Analysis: Glucose contains aldehyde groups, and when heated with freshly prepared Cu(OH)2, it generates brick-red Cu2O precipitate.

43.

Answer: C Analysis: Under acidic conditions, NH_4^+, Al^{3+}, SO_4^{2-}, and Cl-can coexist in large quantities.

44.

Answer: C Analysis: The solid mass reduction of 1.6 g is element O, n (CuO) =0.1 mol, m=8 g.

45.

Answer: C Analysis: ① Preparation of Al(OH)3, ③ removal of Fe, and ④ comparison of acidity are all correct; ② Evaporation of FeCl3 will undergo hydrolysis and cannot yield anhydrous salt.

46.

Answer: A Analysis: Na2SO4 and BaCO3 partially dissolve in water, produce CO2 and completely dissolve when hydrochloric acid is added, and generate CO2 and BaSO4 solid when sulfuric acid is added.

47.

Answer: A Analysis: During discharge, PbO2 gains electrons and undergoes a reduction reaction.

48.